Dual Simplex - Prática

Ex: Resolva o PPL a seguir usando o dual simplex

$$\min z = 6x_1 + 10x_2 + 5x_3$$

s.a.

$$\begin{aligned} x_1 + 2x_2 + 3x_3 &\geq 24 \cdot (-1) \iff -x_1 - 2x_2 - 3x_3 \leq -24 \\ 3x_1 \quad + 2x_3 &\geq 30 \cdot (-1) \quad -3x_1 - 2x_3 \leq -30. \end{aligned}$$

$$x_1, x_2, x_3 \geq 0. \qquad x_1, x_2, x_3 \geq 0.$$

$$\min z = 6x_1 + 10x_2 + 5x_3 \iff z - 6x_1 - 10x_2 - 5x_3 = 0.$$

s.a.

$$\begin{aligned} -x_1 - 2x_2 - 3x_3 + x_4 &= -24 \\ -3x_1 \quad - 2x_3 \quad + x_5 &= -30. \end{aligned}$$

$$x_i \geq 0, \; i = 1, \ldots, 5$$

V.B.	x_1 ↓	x_2	x_3	x_4	x_5	RHS	
z	-6	-10	-5	0	0	0	
x_4	-1	-2	-3	1	0	-24	$\min\left\{\left\|\frac{-6}{-3}\right\|, \left\|\frac{-5}{-2}\right\|\right\} = 2$
← x_5	(-3)	0	-2	0	1	-30	
z	0	-10	-1 ↓	0	-2	60	$L_0^{(1)} = L_0^{(0)} + 6L_2^{(0)}$
← x_4	0	-2	(-7/3)	1	-1/3	-14	$L_1^{(1)} = L_1^{(0)} + L_2^{(1)}$ $\min\left\{\left\|\frac{-10}{-2}\right\|, \left\|\frac{-1}{-7/3}\right\|, \left\|\frac{-2}{-1/3}\right\|\right\} = \frac{3}{7}$
x_1	1	0	2/3	0	-1/3	10	$L_2^{(1)} = -L_2^{(0)}/3$
z	0	-64/7	0	-3/7	-13/7	66	$L_0^{(2)} = L_0^{(1)} + L_1^{(2)}$
x_3	0	6/7	1	-3/7	1/7	6	$L_1^{(2)} = -\frac{3}{7} L_1^{(1)}$
x_1			0			6	$L_2^{(2)} = L_2^{(1)} - 2/3\, L_1^{(2)}$

$$x^* = \begin{bmatrix} 6 \\ 0 \\ 6 \\ 0 \\ 0 \end{bmatrix} \text{ e } z^* = 66$$